

Darukaa.Earth Knowledge Synthesis Report 06

Temperature Extremes, Drought and Vegetation Recovery

Document type: institutional scientific synthesis

Topics: climate, drought, plant_diversity, climate_stress

1. Purpose

This report addresses sites with high temperature, drought, and low or sparse vegetation. The combination is common in degraded drylands and overgrazed or cleared agricultural land.

2. Energy and water at a bare surface

Bare soil absorbs radiation, reaches high surface temperature, and loses water quickly. Sparse vegetation then faces a hotter, drier microclimate than a covered surface would. IPCC and dryland ecology literature describe this as a degradation feedback: vegetation loss begets more vegetation loss.

3. Restoration logic

Revegetation must start with drought-tolerant native species, soil cover (mulch, residue, stones where traditional), and water harvesting. Irrigation-dependent lawns or water-hungry trees are mismatched. If grazing or trampling continues, planting alone will fail.

4. Metrics

- Surface climate stress: downward with cover and shade.
- Soil moisture: upward if infiltration structures work.
- Plant diversity: upward if native assemblages establish.
- Habitat diversity: upward as structure returns.

Time horizon is seasonal for microclimate of mulched plots, and several years for shrub establishment. Woody recovery can take a decade.

5. Source basis

IPCC dryland climate risk; UNCCD restoration principles; ecological literature on nucleation and assisted natural regeneration. Pages refer to this Darukaa document.